E-commerce Customer Behavior Dashboard

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I] Abstract

- **This project focuses on analyzing customer purchasing behavior using a large-scale online retail dataset containing over 500,000 transactions.**
- **The main objective is to identify patterns in customer purchases, revenue trends, and product performance.**
- **The analysis helps in understanding how customers interact with an e-commerce platform.**
- **Data cleaning and preprocessing were carried out using Microsoft Excel to ensure data accuracy and consistency.**
- **Exploratory Data Analysis (EDA) was performed using Python libraries such as Pandas, Matplotlib, and Seaborn.**
- **Key insights were visualized through an interactive Power BI dashboard.**
- **The dashboard enables users to explore data dynamically using filters and visuals.**
- **The project demonstrates how raw transactional data can be transformed into actionable business insights.**
- **These insights can help organizations improve marketing strategies and increase revenue.**

II] Introduction

- **In the modern digital economy, data plays a critical role in business decision-making.**
- **E-commerce platforms generate vast amounts of customer transaction data daily.**
- **Analyzing this data helps businesses understand customer preferences and behavior.**
- **Customer behavior analysis allows companies to improve user experience and retention.**
- **It also helps in identifying trends in purchasing patterns and seasonal demand.**
- **Businesses can use these insights to optimize product offerings and pricing strategies.**
- **This project focuses on analyzing an online retail dataset to uncover such patterns.**
- **The goal is to convert raw data into meaningful and useful information.**
- **Tools like Excel, Python, and Power BI are used to perform this analysis effectively.**

III] Problem Statement

- **Businesses often struggle to understand customer purchasing behavior from raw transactional data.**
- **Large datasets make it difficult to manually identify trends and patterns.**
- **Without proper analysis, companies may miss opportunities to increase revenue.**
- **There is a need to analyze customer data systematically to derive actionable insights.**
- **The project aims to solve this problem by using data analysis techniques.**

Objective Statement:

- **To analyze customer purchasing behavior and extract insights that can improve business decisions, marketing strategies, and revenue growth.**

IV] Objectives

- **To analyze customer purchasing patterns from transaction data**
- **To identify high-value customers and their contribution to revenue**
- **To evaluate top-performing products based on sales**
- **To study revenue trends over time (monthly analysis)**
- **To analyze geographical distribution of sales**
- **To categorize customers based on spending behavior**
- **To create an interactive dashboard for easy data visualization**

V] Dataset Description

- **The dataset used is the Online Retail Dataset from the UCI Machine Learning Repository.**
- **It contains transactional data from a UK-based online retail store.**
- **The dataset covers transactions from December 2010 to December 2011.**
- **It includes over 500,000 records representing individual purchases.**
- **Key Columns:**
- **InvoiceNo – Unique identifier for each transaction**
- **StockCode – Product/item code**
- **Description – Name of the product**
- **Quantity – Number of units purchased**
- **InvoiceDate – Date and time of transaction**
- **UnitPrice – Price per unit**
- **CustomerID – Unique identifier for customers**
- **Country – Customer location**
- **The dataset provides detailed insights into customer behavior and product performance.**

	A	B	C	D	E	F	G	H
1	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
2	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850	United Kingdom
3	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850	United Kingdom
4	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850	United Kingdom
5	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTL	6	12/1/2010 8:26	3.39	17850	United Kingdom
6	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850	United Kingdom
7	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	12/1/2010 8:26	7.65	17850	United Kingdom
8	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	12/1/2010 8:26	4.25	17850	United Kingdom
9	536366	22633	HAND WARMER UNION JACK	6	12/1/2010 8:28	1.85	17850	United Kingdom
10	536366	22632	HAND WARMER RED POLKA DOT	6	12/1/2010 8:28	1.85	17850	United Kingdom
11	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	12/1/2010 8:34	1.69	13047	United Kingdom
12	536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	12/1/2010 8:34	2.1	13047	United Kingdom
13	536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	12/1/2010 8:34	2.1	13047	United Kingdom
14	536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	12/1/2010 8:34	3.75	13047	United Kingdom
15	536367	22310	IVORY KNITTED MUG COSY	6	12/1/2010 8:34	1.65	13047	United Kingdom
16	536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	12/1/2010 8:34	4.25	13047	United Kingdom
17	536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	12/1/2010 8:34	4.95	13047	United Kingdom
18	536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	12/1/2010 8:34	9.95	13047	United Kingdom
19	536367	21754	HOME BUILDING BLOCK WORD	3	12/1/2010 8:34	5.95	13047	United Kingdom
20	536367	21755	LOVE BUILDING BLOCK WORD	3	12/1/2010 8:34	5.95	13047	United Kingdom
21	536367	21777	RECIPE BOX WITH METAL HEART	4	12/1/2010 8:34	7.95	13047	United Kingdom
22	536367	48187	DOORMAT NEW ENGLAND	4	12/1/2010 8:34	7.95	13047	United Kingdom
23	536368	22960	JAM MAKING SET WITH JARS	6	12/1/2010 8:34	4.25	13047	United Kingdom
24	536368	22913	RED COAT RACK PARIS FASHION	3	12/1/2010 8:34	4.95	13047	United Kingdom
25	536368	22912	YELLOW COAT RACK PARIS FASHION	3	12/1/2010 8:34	4.95	13047	United Kingdom
26	536368	22914	BLUE COAT RACK PARIS FASHION	3	12/1/2010 8:34	4.95	13047	United Kingdom

Online Retail

VI] Data Cleaning and Pre-processing

- **Raw data often contains errors, inconsistencies, and missing values.**
- **Data cleaning is essential to ensure accurate analysis.**

Steps performed:

- **Removed cancelled transactions identified by InvoiceNo starting with "C"**
- **Removed negative quantity values representing returns/refunds**
- **Removed rows with missing CustomerID values**
- **Converted InvoiceDate into proper datetime format**
- **Checked for duplicate records and removed them if necessary**
- **Created a new column:**
  - **TotalPrice = Quantity × UnitPrice**
- **These steps improved data quality and ensured reliable analysis results.**

	A	B	C	D	E	F	G	H	I	J	K
1	InvoiceN	StockCo	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	TotalPric	Year	Month
2	536365	85123A	WHITE HANGING H	6.00	12/1/2010	2.55	17850	United Kingdom	15.3	2010	Dec
3	536365	71053	WHITE METAL LAN	6.00	12/1/2010	3.39	17850	United Kingdom	20.34	2010	Dec
4	536365	84406B	CREAM CUPID HEA	8.00	12/1/2010	2.75	17850	United Kingdom	22	2010	Dec
5	536365	84029G	KNITTED UNION FL	6.00	12/1/2010	3.39	17850	United Kingdom	20.34	2010	Dec
6	536365	84029E	RED WOOLLY HOTT	6.00	12/1/2010	3.39	17850	United Kingdom	20.34	2010	Dec
7	536365	22752	SET 7 BABUSHKA N	2.00	12/1/2010	7.65	17850	United Kingdom	15.3	2010	Dec
8	536365	21730	GLASS STAR FROST	6.00	12/1/2010	4.25	17850	United Kingdom	25.5	2010	Dec
9	536366	22633	HAND WARMER UN	6.00	12/1/2010	1.85	17850	United Kingdom	11.1	2010	Dec
10	536366	22632	HAND WARMER RE	6.00	12/1/2010	1.85	17850	United Kingdom	11.1	2010	Dec
11	536367	84879	ASSORTED COLOUF	32.00	12/1/2010	1.69	13047	United Kingdom	54.08	2010	Dec
12	536367	22745	POPPY'S PLAYHOUS	6.00	12/1/2010	2.10	13047	United Kingdom	12.6	2010	Dec
13	536367	22748	POPPY'S PLAYHOUS	6.00	12/1/2010	2.10	13047	United Kingdom	12.6	2010	Dec
14	536367	22749	FELTCRAFT PRINCE	8.00	12/1/2010	3.75	13047	United Kingdom	30	2010	Dec
15	536367	22310	IVORY KNITTED MU	6.00	12/1/2010	1.65	13047	United Kingdom	9.9	2010	Dec
16	536367	84969	BOX OF 6 ASSORTE	6.00	12/1/2010	4.25	13047	United Kingdom	25.5	2010	Dec
17	536367	22623	BOX OF VINTAGE JI	3.00	12/1/2010	4.95	13047	United Kingdom	14.85	2010	Dec
18	536367	22622	BOX OF VINTAGE A	2.00	12/1/2010	9.95	13047	United Kingdom	19.9	2010	Dec
19	536367	21754	HOME BUILDING BL	3.00	12/1/2010	5.95	13047	United Kingdom	17.85	2010	Dec
20	536367	21755	LOVE BUILDING BL	3.00	12/1/2010	5.95	13047	United Kingdom	17.85	2010	Dec
21	536367	21777	RECIPE BOX WITH	4.00	12/1/2010	7.95	13047	United Kingdom	31.8	2010	Dec
22	536367	48187	DOORMAT NEW EN	4.00	12/1/2010	7.95	13047	United Kingdom	31.8	2010	Dec
23	536368	22960	JAM MAKING SET V	6.00	12/1/2010	4.25	13047	United Kingdom	25.5	2010	Dec

```python
df = df[~df['InvoiceNo'].astype(str).str.startswith('C')]
df = df[df['Quantity'] > 0]
df = df.dropna(subset=['CustomerID'])
df['TotalPrice'] = df['Quantity'] * df['UnitPrice']
```

[3] Python

```python
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])

df['Year'] = df['InvoiceDate'].dt.year
df['Month'] = df['InvoiceDate'].dt.month
df['MonthName'] = df['InvoiceDate'].dt.strftime('%b')
```

[4] Python

VII] Exploratory Data Analysis (Python)

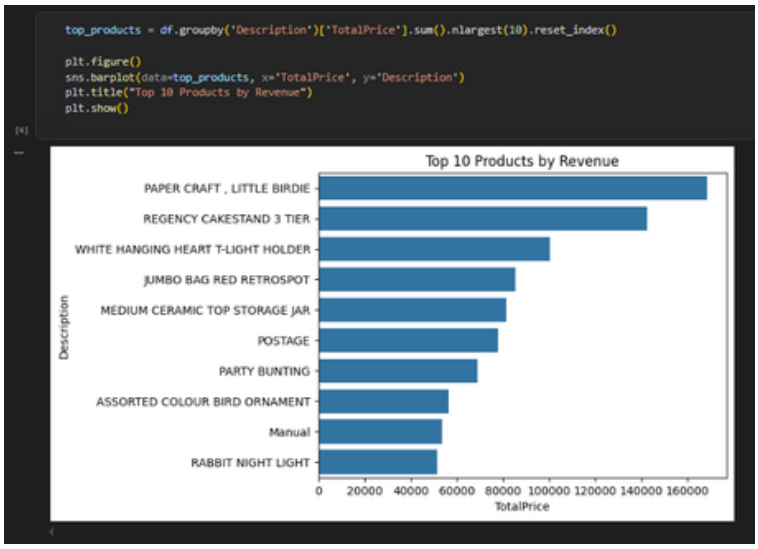
- **Exploratory Data Analysis helps in understanding the dataset before building dashboards.**
- **Python libraries such as Pandas, Matplotlib, and Seaborn were used.**

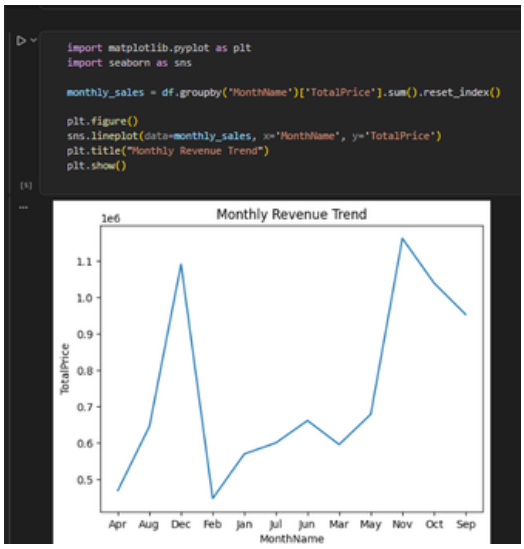
Analysis performed:

- **Generated summary statistics of numerical columns**
- **Analyzed monthly revenue trends**
- **Identified top-selling products**
- **Examined revenue distribution across countries**
- **Checked customer purchase frequency**

Observations:

- **Revenue increases significantly towards the end of the year**
- **Few products contribute heavily to total revenue**
- **Sales are concentrated in specific regions**
- **Visualizations were created to support these findings.**





VIII] Dashboard Development (Power BI)

- **Power BI was used to create an interactive and visually appealing dashboard.**
- **The cleaned dataset was imported into Power BI.**
- **Measures and calculated fields were created for analysis.**

Dashboard Components:

KPIs:

- **Total Revenue**
- **Total Orders**
- **Total Customers**
- **Average Order Value**

Visualizations:

- **Monthly Revenue Trend (Line Chart)**
- **Top 10 Products (Bar Chart)**
- **Revenue by Country (Map)**
- **Customer Segmentation (Donut Chart)**
- **Revenue vs Orders (Combo Chart)**

Filters:

- **Country filter**
- **Month filter**
- **The dashboard allows users to interact with data dynamically.**

IX] Key Insights

- **The United Kingdom contributes the highest share of total revenue**
- **Sales show a strong increase during November and December**
- **A small percentage of customers generate a large portion of revenue**
- **Most customers fall into the low spending category**
- **Certain products dominate overall sales performance**
- **Seasonal trends play a major role in revenue fluctuations**

X] Business Recommendations

- **Focus marketing campaigns on high-value customers**
- **Increase promotional efforts during peak sales months**
- **Highlight top-performing products to maximize sales**
- **Expand business to other regions to diversify revenue sources**
- **Use customer segmentation for targeted marketing strategies**

XI] Conclusion

- **The project successfully analyzed customer purchasing behavior using real-world data.**
- **Data cleaning and preprocessing ensured accurate analysis.**
- **Exploratory analysis helped uncover important patterns in the dataset.**
- **The Power BI dashboard provided clear and interactive visualizations.**
- **The insights derived can help businesses improve decision-making.**
- **This project demonstrates the importance of data analysis in modern business environments.**

XII] Tools & Technologies

- **Microsoft Excel – Data cleaning and preprocessing**
- **Python (Pandas, Matplotlib, Seaborn) – Data analysis and visualization**
- **Power BI – Dashboard creation and visualization**

XIII] Limitations

- **Dataset does not include customer demographic details**
- **Limited to historical transactional data only**
- **No predictive or machine learning models applied**

XIV] Future Scope

- **Implement machine learning models for customer segmentation**
- **Predict customer lifetime value**
- **Build recommendation systems for products**
- **Integrate real-time data for live dashboards**

XV] References

- **UCI Machine Learning Repository – Online Retail Dataset**
- **Power BI Official Documentation**
- **Python Libraries Documentation (Pandas, Seaborn, Matplotlib)**